

Assignment - 2

20/07/26

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1) Design a Machine Learning Model for spam Email classification.

a) choice of Algorithm

A Suitable Algorithm for spam email classification is

Naive Bayes.

Reason for choosing Naive Bayes:-

- works well with text data.
- Fast & efficient for large email datasets.
- Handles high-dimensional features effectively.
- Based on probability, making it suitable for word-frequency analysis.

Other alternatives:-

- Support Vector Machine (SVM):- High accuracy but computationally Expensive.

• Logistic Regression :- simple, Interpretable, & effective for binary classification.

b) Preprocessing steps

1. Data collection

- collect labeled emails as spam or ham

2. Tokenization

- split email text into individual words.

3. Lowercasing

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- Convert all text to lowercase.

4. Stop-word Removal

- Remove common words such as is, the, and, of.

5. Stemming / Lemmatization

- Reduce words to their root form.

- Example: running → run

6. Vectorization

- Convert text into numerical features using:

- Bag of Words (BoW)

- TF-IDF (Term Frequency - Inverse Document Frequency)

7. Model Training

- Train the Naive Bayes Classifier using the processed data

8. Prediction:-

- Classify new emails as spam or Ham

C) Evaluation Metrics

Metric

Formula

Purpose

Accuracy

$$(TP+TN)/(TP+TN+FP+FN)$$

Overall correctness of the model

Precision

$$TP/(TP+FP)$$

Measures how many predicted spam emails are actually spam

Recall

$$TP/(TP+FN)$$

Measures how many actual spam emails are correctly detected

F1-Score

$$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Balances precision &

conclusion:-

A good spam classifier should achieve high precision & high recall.

2) Candidate Elimination Algorithm.

a) Initialization of S and G

Given attributes

sky, AirTemp, Humidity, wind, water, Forecast

Target Concept :- Enjoysport

Initially:-

• specific Boundary (S):

- Most specific hypothesis

- $S = \{(\phi, \phi, \phi, \phi, \phi, \phi)\}$

• General Boundary (G):

- Most general hypothesis

- $G = \{(? , ? , ? , ? , ? , ?)\}$

where:

- ϕ = No value specified

- ? = Any value

b) updating S and G

consider the following Training Examples:-

Example	sky	AirTemp	humidity	wind	water	Forecast	Enjoysport
1	sunny	warm	Normal	Strong	warm	same	yes
2	sunny	warm	High	strong	warm	same	yes
3	Rainy	cold	High	Strong	warm	change	No

Step 1 (positive Example)

Example:-

<Sunny, warm, Normal, strong, warm, same>

update S

S becomes:

<sunny, warm, Normal, strong, warm, same>

G remains:

<?, ?, ?, ?, ?, ?>

Step 2 (positive Example)

Example :-

<Sunny, warm, High, strong, warm, same>

Generalize S where attributes differ.

updated S:-

<Sunny, warm, ?, strong, warm, same>

G remains unchanged:

<?, ?, ?, ?, ?, ?>

Step 3 (Negative Example)

Example:-

<Rainy, Cold, High, strong, warm, change>

S remains:-

<Sunny, warm, ?, strong, warm, same>

Specialize G to exclude the negative example while

remaining consistent with S.

possible G:

< sunny, ?, ?, ?, ?, ? >

< ?, warm, ?, ?, ?, ? >

< ?, ?, ?, ?, ?, same >

c) Candidate - Elimination vs Find-S

Feature	candidate - Elimination	Find-S
		yes
uses positive examples	yes	NO
uses negative examples	yes	onlys
Maintains hypothesis boundaries	yes	NO
Finds all consistent hypotheses	Yes	Limited
Generalization capability	High	
Accuracy	Better because it considers both positive & negative Examples.	lower, as it ignores negative Example

conclusion:-

The Candidate - Elimination Algorithm is more powerful than the Find-S Algorithm because it uses both positive & negative examples to maintain the complete version space of consistent hypotheses, leading to better generalization.